

CAHIER DES CHARGES

Specifications Document

FLIPPER 42

Portable Multi-Protocol Security Research Device

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Authors	FLIPPER 42 Team - 42 Paris

Table of Contents

1. Introduction.....	3
1.1 Project Overview.....	3
1.2 Scope.....	3
1.3 Definitions & Abbreviations.....	3
2. Functional Requirements.....	4
2.1 Core Functions.....	4
2.2 Sub-GHz (CC1101).....	4
2.3 NFC (ST25R3916).....	4
2.4 RFID (EM4095).....	4
2.5 User Interface.....	4
3. Hardware Architecture.....	5
3.1 System Block Diagram Overview.....	5
3.2 Microcontroller - ATmega2560.....	5
3.3 Communication Buses.....	6
3.4 Power Supply.....	6
4. Non-Functional Requirements.....	7
4.1 Physical.....	7
4.2 Electrical.....	7
4.3 Software & Firmware.....	7
4.4 Safety.....	7
5. Bill of Materials (BoM).....	7
6. Implementation Procedure.....	8
6.1 Development Phases.....	8
7. Acceptance Criteria & Test Plan.....	8
7.1 Functional Tests.....	8
7.2 Compliance Checks.....	9
8. References & Datasheets.....	9
9. Notes.....	9
9.1 Hardware Block Diagram Description.....	9

1. Introduction

1.1 Project Overview

The FLIPPER 42 is a portable, open-hardware multifunctional device designed for interaction with various electronic access-control and radio-frequency systems. Inspired by the Flipper Zero, the project explores low-level embedded systems design and RF security research using an alternative hardware architecture built around the ATmega2560 microcontroller.

The device is conceived for:

- Educational purposes and academic training in embedded systems
- Security research and ethical hacking activities
- Reading, cloning, and replaying NFC/RFID credentials on authorised systems
- Sub-GHz radio signal acquisition and controlled reproduction
- Hands-on exploration of hardware security protocols

1.2 Scope

This document describes the technical and functional specifications of the FLIPPER 42 hardware platform, covering architecture, communication modules, power supply, firmware behaviour, and acceptance criteria. It serves as the reference baseline for the 42 Paris electronics module validation.

1.3 Definitions & Abbreviations

Acronym / Term	Definition
CDC	Cahier des Charges
MCU	Microcontroller Unit
RFID	Radio-Frequency IDentification
NFC	Near-Field Communication
Sub-GHz	Radio frequencies below 1 GHz (e.g. 433 MHz, 868 MHz)
SPI	Serial Peripheral Interface - synchronous serial bus
I2C	Inter-Integrated Circuit - two-wire serial bus
UART	Universal Asynchronous Receiver-Transmitter
BMS	Battery Management System
ADC	Analog-to-Digital Converter
GPIO	General-Purpose Input/Output
EEPROM	Electrically Erasable Programmable Read-Only Memory
LF	Low Frequency (125 kHz RFID band)
HF	High Frequency (13.56 MHz NFC band)

2. Functional Requirements

2.1 Core Functions

The device shall implement the following primary functions:

1. RF Signal Detection & Identification - detect and classify incoming radio signals in supported bands
2. NFC Credential Reading - read and decode ISO 14443 and ISO 15693 compliant tags at 13.56 MHz
3. RFID Credential Reading - read 125 kHz LF tags using the EM4095 transceiver
4. Data Storage - persist captured payloads, signal recordings, and configurations on microSD
5. Signal Replay - reproduce previously recorded signals upon authorised user action
6. External Expansion - expose hardware interfaces (SPI, UART, GPIO) for optional add-on modules

2.2 Sub-GHz (CC1101)

The CC1101 low-power Sub-1 GHz RF transceiver shall support:

- Single-band operation dedicated exclusively to the 433 MHz ISM band
- Serial modes for reading, capturing, and replaying raw RF frames
- A path to a physical coaxial connector for an external 433 MHz antenna
- Communication with the MCU via a 4-wire SPI interface at 3.3 V

2.3 NFC (ST25R3916)

The ST25R3916B-AQWT high-performance NFC frontend shall support:

- Reader/Writer mode for ISO 14443-A/B and ISO 15693 tagsAlso @
- Card emulation mode (Type A / Type B)
- Reading, storing, and replaying NDEF messages and raw tag dumps
- Communication with the MCU via SPI at 3.3 V

2.4 RFID (EM4095)

The EM4095 125 kHz RFID transceiver shall:

- Read EM4100-compatible and compatible LF tags
- Provide demodulated data via GPIO/UART to the MCU
- Store captured UIDs and data payloads to microSD

2.5 User Interface

The device shall provide a 6-button navigation interface:

Button	Direction	Function
◀	Left	Navigate left / back
▼	Down	Scroll down / decrease value
•	Select	Confirm / enter
▲	Up	Scroll up / increase value
▶	Right	Navigate right / forward
↶	Outside / Isolated	Back / Return

An OLED display (connected via I2C) shall provide a menu-driven interface showing:

- Main menu with module selection
- Real-time signal activity and status
- Stored file browser for saved captures
- Battery level and system status

3. Hardware Architecture

3.1 System Block Diagram Overview

The hardware architecture is centred on the ATmega2560 MCU and interconnects the following subsystems via dedicated bus protocols as described in the uploaded block diagram:

Subsystem	Interface	Voltage	Notes
MCU - ATmega2560	Central	5 V	Main controller, 256 KB Flash, 8 KB SRAM, 4 KB EEPROM
OLED Display	I2C	3.3/5V	ER-OLED015-2Y, 128×64 px
microSD Card	SPI	3.3 V	FAT32 formatted; stores captures & config
CC1101 Sub-GHz Transceiver	SPI	3.3 V	Level-shifted; 433 MHz only; external antenna
ST25R3916 NFC Frontend	SPI	3.3 V	ISO 14443/15693 reader + card emulation
EM4095 RFID Transceiver	GPIO	5 V	125 kHz LF RFID
Battery + BMS	Power	3.7 V	LiPo cell with protection circuit
USB / UART Bridge	UART	5 V	PC connectivity, firmware upload
Buttons (5×)	GPIO	3.3 V	Navigation; debounced in firmware
Level Shifter	Logic	3.3/5V	Bidirectional; protects 3.3 V peripherals

Voltage Divider / ADC	Analog	Var.	Battery voltage sense via MCU ADC
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3.2 Microcontroller - ATmega2560

The ATmega2560 (TQFP-100) is the central processing unit. Key specifications:

Parameter	Value
Architecture	8-bit AVR RISC
Clock speed	Up to 16 MHz (external crystal)
Flash memory	256 KB (in-system programmable)
SRAM	8 KB
EEPROM	4 KB
GPIO	86 pins
SPI interfaces	1× hardware SPI (shared with CC1101, SD, NFC)
I2C (TWI)	1× hardware TWI (OLED)
UART	4× hardware UART (1 used for USB bridge)
ADC	16-channel, 10-bit
Timers	6× (8-bit and 16-bit)
Operating voltage	4.5 V – 5.5 V

3.3 Communication Buses

3.3.1 SPI Bus (3.3 V)

A single SPI bus (MOSI, MISO, SCK) serves the CC1101, microSD, and ST25R3916. Each peripheral is independently selected via a dedicated active-low CS (chip select) line driven by the MCU GPIO. The bus operates at 3.3 V logic; a bidirectional level shifter protects all 3.3 V devices from the 5 V MCU outputs.

3.3.2 I2C Bus

The I2C bus connects the OLED display to the MCU using standard 400 kHz fast-mode. Pull-up resistors (4.7 kΩ typical) shall be fitted on SDA and SCL lines.

3.3.3 UART

UART0 is routed to a USB-to-UART bridge IC (e.g. CH340G or CP2102) for PC connectivity. This enables firmware flashing via avrdude and runtime serial debugging at 115200 baud.

3.4 Power Supply

Domain	Source	Regulation	Consumers
5 V main	USB or LiPo via boost	LDO / boost converter	MCU, EM4095, USB bridge

3.3 V rail	AMS1117-3.3 or equiv.	LDO from 5 V	CC1101, NFC, OLED, SD, level shifter
Battery	LiPo 3.7 V / 1000 mAh+	BMS + protection IC	System supply when off USB

The BMS shall provide over-charge, over-discharge, and short-circuit protection. A voltage divider on the battery line feeds the MCU ADC for state-of-charge estimation.

4. Non-Functional Requirements

4.1 Physical

Requirement	Target
Weight	TODO < 400g
Enclosure	PCB-mount shell Initially then 3D-printed
Connector	USB-C or Micro-USB for power and programming

4.2 Electrical

Requirement	Target
Supply input	5 V via USB; 3.7 V LiPo battery
3.3 V rail	+5% tolerance
Autonomy	TODO

4.3 Software & Firmware

- Firmware written in C using the “avr” library toolchain
- Menu-driven OLED interface with 5-button navigation
- FAT32 filesystem on microSD
- UART debug console at 115200 baud accessible via USB

4.4 Safety

The FLIPPER 42 shall be operated exclusively in compliance with applicable radio regulations.

The device is intended for educational and security research only.

5. Bill of Materials (BoM)

The following table lists the principal components identified for the FLIPPER 42 prototype. Quantities are for a single unit.

Qty	Component	Part Number	Interface	Notes
1	MCU	ATmega2560-16AU	SPI/I2C/UART/ GPIO	Main controller, TQFP-100
1	Sub-GHz Transceiver	CC1101	SPI 3.3 V	433 MHz; external antenna
1	NFC Frontend	ST25R3916B-AQWT	SPI 3.3 V	ISO 14443 / 15693
1	RFID Transceiver	EM4095	GPIO 5 V	125 kHz LF RFID
1	OLED Display	ER-OLED015-2	I2C	128×64 px

1	microSD Socket	1040310811 (Molex)	SPI 3.3 V	FAT32; micro-SD
1	3.3 V LDO	AMS1117-3.3	Power	300 mA min.
1	Battery BMS		Power	LiPo protection
1	LiPo Battery	3.7 V / 1000+ mAh	Power	Rechargeable
1	Coaxial Antenna	433 MHz SMA	RF	External removable
5	Tactile Switches	Generic 6×6 mm	GPIO	Navigation buttons
--	Passives (R, C, L)	Various	PCB	Per datasheet values

6. Implementation Procedure

6.1 Development Phases

Phase	Activity	Deliverable
1 - Design	Schematic capture, component selection, BOM validation	Schematic PDF, BoM spreadsheet
2 - PCB Layout	Routing, DRC KiCad Files	PCB fabrication order
3 - Prototyping	Breadboard / protoboard assembly and initial power-on	Powered prototype, voltage checks
4 - Firmware	Development per module (CC1101, NFC, RFID, SD)	Firmware repo, unit tests
5 - Integration	System integration, module interaction, OLED UI	Working integrated prototype
6 - Validation	Functional test against acceptance criteria	Test report
7 - Documentation	CDC finalisation, code documentation, README	Final CDC PDF, GitHub

7. Acceptance Criteria & Test Plan

7.1 Functional Tests

Test ID	Module	Procedure	Expected Result	Pass Criteria
T-01	Power	Apply 5 V USB; measure 5 V and 3.3 V rails	Both rails within $\pm 5\%$	Voltmeter reading in spec
T-02	MCU	Flash blink firmware; observe LED	LED blinks at 1 Hz	Visible blink
T-03	OLED	Flash display test; observe screen	All pixels lit, menu displayed	Visual Confirmation
T-04	SD Card	Write and read test file via firmware	File written and read back correctly	Data integrity 100%
T-05	CC1101	Capture 433 MHz remote signal; display capture	Signal captured and stored on SD	Valid file on SD card

T-06	CC1101	Replay stored 433 MHz signal to target device	Target device responds	Repeatable successful replay
T-07	NFC	Read ISO 14443-A tag; display UID on OLED	UID displayed correctly	Matches reference reader
T-08	RFID	Read EM4100 125 kHz tag; display data	Tag data displayed correctly	Matches reference reader
T-09	Battery	Operate 4 h on battery; monitor voltage	Device operational for ≥ 4 h	No shutdown under load
T-10	Buttons	Press each button; verify OLED response	Correct navigation action per button	All 6 buttons functional

7.2 Compliance Checks

- All RF transmissions verified to comply with 433 MHz ISM band limits
- Device shall not transmit on unauthorised frequencies

8. References & Datasheets

Document	Source
ATmega2560 Datasheet	https://www.microchip.com/en-us/product/atmega2560
CC1101 Datasheet	https://www.ti.com/lit/ds/symlink/cc1101.pdf?ts=1780550466112
ST25R3916B Datasheet	https://www.st.com/resource/en/datasheet/st25r3916b.pdf
EM4095 Datasheet	https://www.emmicroelectronic.com/sites/default/files/products/datasheets/em4095_ds.pdf
microSD Socket (Molex)	Part 1040310811
Project GitHub Repository	https://github.com/youpaw/elec-module
Flipper Zero Reference	https://flipperzero.one

9. Notes

9.1 Hardware Block Diagram Description

The following block diagram shows the interconnections for the FLIPPER 42 system.

